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Threaded insert for an airfield light fixture

Abstract

A threaded insert for an air field light fixture. The threaded insert is manufactured to repair or replace the threading in such light fixtures by providing the airfield light fixture with a replacement threaded insert to permit quick and simple repair upon the occurrence of a broken bolt or stripped threading.

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Background/Summary

RELATED APPLICATIONS AND INCORPORATION BY REFERENCE (1) This application claims benefit and priority to U.S. Provisional patent application Ser. No. 63/441,522, filed Jan. 27, 2023. The foregoing application is hereby incorporated by reference.

FIELD OF DISCLOSURE

(1) The present disclosure generally relates to a threaded insert for an airfield light fixture.

BACKGROUND

(2) Inset light fixtures have long been used at airports to provide lighting for both airport runways and taxiways. Typically each light fixture includes a container which houses electrical components and is embedded into the concrete. Typically each light fixture includes a cover which is mountable to the top of the container with plurality of bolts which extend into threaded openings in the container.

(3) Due to its exposure to the elements, the light fixture and its corresponding bolts undergo a significant amount of wear and tear. Bolts and light fixtures often rust or deteriorate in the normal course of operation. In addition, maintenance of these light fixtures requires the removal of the bolts in order to gain access to the container and electrical components. A common problem with these light fixtures is that the bolts rust and break within the threaded opening of the container or the bolts strip the threading upon their removal.

(4) In such case, the threaded openings of the container are blocked and/or damaged and require repair before the light fixture can be resecured within the pavement. This often requires broken bolts to be removed and damaged threadings be rethreaded on site.

(5) U.S. Pat. No. 6,550,931 discloses a threaded insert for use with an airfield light fixture. However, this patent discloses removable inserts that are used in connection with each and every bolt of the device. This prior reference discloses a base can which specifically requires the inserts in the disclosure in order to function. The disclosure of the '931 patent is not directed toward an insert that can replace any stripped threading in airfield light fixtures. Instead the '931 Patent is directed to inserts which are specially created for the disclosed light fixture.

(6) A long-felt, but unmet, need exists for a simple and easy way to re-thread or repair the threaded holes within any air field light fixture container.

SUMMARY OF DISCLOSURE

(7) This summary is provided to introduce a selection of concepts in a simplified form further described in the detailed description of the disclosure. This summary is not intended to identify key or essential inventive concepts of the claimed subject matter, nor is it intended for determining the scope of the claimed subject matter.

(8) A long-felt, but unmet, need exists for a simple and easy method and device to re-thread and repair damaged threadings within an airfield light fixture.

(9) According to the present invention, this object is achieved by providing the airfield light fixture with a replacement threaded insert to permit quick and simple repair upon the occurrence of a broken bolt or stripped threading.

(10) Embodiments of the present disclosure address one or more of the disadvantages and problems discussed above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a perspective view of the threaded insert.

(2) FIG. 2 depicts a second perspective view of the threaded insert.

(3) FIG. 3 depicts a side view of the threaded insert.

(4) FIG. 4 depicts a top view of the threaded insert.

(5) FIG. 5 depicts a second side view of the threaded insert.

(6) FIG. 6a depicts the container portion of a light fixture base can embedded in pavement.

(7) FIG. 6b depicts a close up view of a portion of the light fixture base can of FIG. 6a.

(8) FIG. 7 depicts the container and cover of the airfield light fixture bolted into the pavement.

(9) FIG. 8 depicts an exploded view of the airfield light fixture, a bolt, and threaded insert.

(10) FIG. 9 depicts a flow diagram of the steps of using the threaded insert.

DETAILED DESCRIPTION

(11) This disclosure as a whole may be best understood by reference to the following detailed description when read in conjunction with the accompanying drawings, drawing descriptions, abstract, background, field of disclosure, and associated headings. Identical reference numbers when found on different figures identify the same elements or a functionality equivalent element. The elements listed in the abstract are not referenced but nevertheless refer by association to the elements of the detailed description and associated disclosure.

The Insert

(12) FIGS. 1-5 depict a threaded insert **100**. The insert **100** has a top end **102**, bottom end **104**, and side wall **106** extending between the top end **102** and bottom end **104**. The insert **100** is a generally cylindrical member with a hole **108** extending longitudinally therethrough. In an embodiment, the hole **108** includes an internal threading. In alternative embodiment, the hole **108** does not comprise internal threading.

(13) As depicted in FIGS. 1-5, a lip **116** projects radially and outwardly from the bottom end **104** of the insert **100**. In the preferred embodiment, the lip **116** is integral with and comprised of the same material as the sidewall **106**.

(14) As depicted in FIGS. 1-5, the top end **102** of the insert **100** is tapered **110**, which assists with installing the insert **100** into the airfield light fixture **200**. In an alternative embodiment, the top end **102** of the inset does not include a tapered portion. A plurality of protrusions **114** extend radially from the exterior side wall **106** of the insert **100** which assists with securing the insert **100** into place. In the embodiment depicted in FIGS. 1-5, the protrusions **114** comprise a plurality of splines.

(15) In the preferred embodiment, and as depicted in FIG. 1, the interior side wall **112** of the insert **100** is threaded to receive bolts and secure the cover of the airfield light fixture to the container of the airfield light fixture.

(16) In a preferred embodiment, the tapered portion **110** forms a five degree angle with the side wall **106** of the insert **100**. In the preferred embodiment, the tapered portion **110** includes a plurality of protrusions **114** of similar shape as the protrusions **114** extending from the exterior side wall **106**. In the preferred embodiment, the tapered portion is about 0.0750 in. in length. In alternative embodiments, other configurations of the tapered portion are considered.

(17) The insert **100** may be manufactured to comply with an airfield light container **200** fixture having holes of any size. In a first embodiment of the insert **100**, the insert has a length of about 0.4550 in. from the lip **116** to the top end **102**, an outside sidewall **106** diameter of about 0.5 in. (without the protrusions **114**), a protrusion extending the length of up to 0.036 in. away from the side wall **106**, and an interior sidewall **106** diameter of about 0.375 in. In this embodiment, the diameter of the lip **116** is approximately 0.6875 in. and has a thickness of about 0.0800 in.

(18) In a second embodiment of the insert **100**, the insert has a length of about 0.8300 in. from the lip **116** to the top end **102**, an outside sidewall **106** diameter of about 0.5 in. (without the protrusions **114**), a protrusion extending the length of up to 0.036 in., and an interior sidewall thickness of about 0.375 in. In an alternative embodiment, the diameter of the lip **116** is approximately 0.6875 in. and has a thickness of about 0.0787 in.

(19) In another embodiment, the outside sidewall **106** has a diameter of about 10 mm. Other embodiments are considered which allow for universal installation of the insert **100** into airfield light fixtures **200**.

(20) The insert **100** of the present disclosure is comprised of thermoplastics, composites, metals, or the like (including but not limited to Carbon Steel and/or Stainless Steel). Many inserts currently in use in the art are made from Carbon Steel. In a preferred embodiment, the insert **100** of the present disclosure is made from **316** Stainless Steel.

(21) Inserts currently in use with airfield light fixtures do not include a tapered barrel. In the

preferred embodiment of the present disclosure, the barrel of the insert **100** is tapered to assist with drilling the insert **100** into the container of the airfield light fixture **200**.

(22) Inserts currently in use in the art typically have an exterior sidewall diameter of 17/32" (without protrusions). In an embodiment of the present disclosure, the insert **100** has an exterior sidewall thickness of 1/2" (excluding protrusions **114**). This provides additional space for protrusions **114** and allows the inserts **100** to include protrusions **114** with longer splines. In effect, the longer splines prevent stripping and lock the insert **100** into the airfield light fixture **200** more effectively.

(23) In the preferred embodiment of the present disclosure, the protrusions **114** extend up to 0.036 in. from the exterior sidewall **106** of the insert **100**. The protrusions **114** of the present disclosure cut deeper and are spread wider apart than the inserts **100** in the prior art. The insert **100** of the present disclosure locks more effectively into the airfield light fixture **200** and prevents rotation of the insert **100** when torquing a bolt. The insert **100** of the present disclosure also prevents stripping or destruction of the hole **210** in the airfield light fixture **200** (i.e., it prevents a user from making the hole **210** larger than the insert **100** itself during installation).

The Airfield Light Fixture

(24) FIGS. **6** and **7** depict a standard airfield light fixture **200**. An airfield light fixture **200** is typically installed in the ground **202** such as the pavement of an airfield runway. The light fixture **200** generally comprises a base can **204** with a container **206** and flange **208** portion. The container portion **206** includes a generally cylindrical side wall with an upper and lower end. A flange **208** is mounted on the upper end of the side wall and extends outwardly and radially therefrom. A plurality of threaded holes **210** are spaced apart along the perimeter of the flange **208**, the holes **210** are often designed with threading **216** to receive a standard bolt **214**.

(25) Each base can **204** has a corresponding cover **212** which is generally circular in shape and sized to cover the open upper end of the base can **204**. The cover **212** includes a plurality of holes positioned for alignment with the threaded holes **210** in the base can **204**.

(26) The cover **212** is secured to the base can **204** by aligning the holes of the cover and the holes **210** and securing said bolts **214** into the holes **210**.

Method of Use

(27) FIG. **8** depicts an exploded view of the airfield light fixture, a bolt, and threaded insert. FIG. **9** depicts a flow diagram of the method of using the threaded inserts **100** of the present disclosure **900**. Over time, the threading **216** within the holes **210** may become stripped or damaged. The threaded inserts **100** allow a user to repair or replace the threading within a light fixture **200** quickly and easily.

(28) In a first step **902**, a user may remove the bolts **214** from the holes **210** of the airfield light fixture. In a second step **904**, a user removes the cover/lid **212** from the top of the base can **204** of the airfield light fixture. The base can **204** is removable from the ground and is not fixed to the airfield runway **202**. Once the lid **212** is removed, a user can access the interior of the airfield light fixture to make any necessary repairs to the light fixture itself or to the electrical components therein. In a third step, the user inspects the holes **210** to determine whether the threading **216** has been damaged.

(29) If the threading has been damaged, the user proceeds to a fourth step **908** and drills a threaded insert **100** into any holes **210** having damaged threads **216**. In a preferred embodiment, the threaded insert **100** is drilled upward from the bottom of each hole **210**. In a fifth step **910**, the user places the cover/lid **212** on top of the base can **204** of the light fixture. In a final step **912**, the user can reapply the bolts **214** to the holes **210** (at least some of which now include the threaded inserts **100**) to secure the light fixture in place **912**.

(30) If the threading **216** within the holes **210** has not been damaged or stripped, the user may elect to simply reattach the lid to the light fixture **910** and refasten the bolts **214** to the holes **210**. In alternative embodiments, even if the threading **216** of the holes **210** has not been damaged, a user

may elect to apply the threaded inserts **100** to the holes.

(31) The walls of the threaded insert **106** include a plurality of protrusions **114** (i.e., ribbed splines). The walls of the threaded insert **106** (with the ribbed splines) are slightly larger in diameter than the holes **210**. When the insert is drilled into the hole **210**, the splines cut into the **208** and lock the insert **100** in place. Once the insert **100** is in place, a user can secure a bolt **214** within the interior threading **112** of the insert **100**. The lip **116** of the insert **100** contacts the lower face of the flange **208** to prevent the insert from being removed through the upper face of the flange **208** if the user later tries to remove the bolt **214**.

(32) It is understood that the preceding is merely a detailed description of some examples and embodiments of the present invention and that numerous changes to the disclosed embodiments may be made in accordance with the disclosure made herein without departing from the spirit or scope of the invention. The preceding description, therefore, is not meant to limit the scope of the invention, but to provide sufficient disclosure to allow one of ordinary skill in the art to practice the invention without undue burden.

Claims

1. A threaded insert for an airfield light fixture comprising: a top end and a bottom end; a cylindrical wall running between the top and bottom ends of the threaded insert; a lip integral with the cylindrical wall proximate the bottom end of the threaded insert, wherein the lip has a diameter greater than the diameter of the cylindrical wall; an opening proximate the top end of the threaded insert and extending throughout the cylindrical wall to form a cavity between the interior of the cylindrical wall and the lip; a plurality of protrusions extending radially from the exterior of the cylindrical wall; and a tapered portion proximate the end of the cylindrical wall opposite the lip, wherein the cavity extending throughout the cylindrical wall and tapered portion has a substantially uniform radius and the exterior of the tapered portion forms an angle with the plane of the cylindrical wall; and wherein the interior of the cylindrical wall is threaded.
2. The threaded insert of claim 1, wherein the plurality of protrusions extending radially from the exterior of the cylindrical wall comprise a plurality of splines, wherein a base portion of each of the plurality of splines integral with the cylindrical wall is greater in thickness than a top end of each of the plurality of splines extending from the cylindrical wall.
3. The threaded insert of claim 2, wherein the plurality of splines extend up to approximately 0.036 in. from the cylindrical wall.
4. The threaded insert of claim 1, wherein the tapered portion forms between a two to six degree angle with the plane of the cylindrical wall.
5. The threaded insert of claim 1, wherein the tapered portion forms a five degree angle with the plane of the cylindrical wall.
6. The threaded insert of claim 1, wherein the diameter of the cylindrical wall is approximately 0.5 in.
7. The threaded insert of claim 1, wherein the length from the top end of the threaded insert to the bottom end of the threaded insert is approximately between 0.455 in. to 0.83 in.
8. The threaded insert of claim 1, wherein the lip comprises a diameter of approximately 0.6875 in. and a thickness of approximately between 0.0787 in. to 0.08 in.
9. The threaded insert of claim 1, wherein the cavity between the interior of the cylindrical wall and the lip has a diameter of approximately 0.375 in.
10. An airfield light fixture apparatus comprising: a cylindrical light fixture base can manufactured to be embedded into an airfield runway, the base can comprising a container portion to be embedded in the ground and a flange portion; the container portion comprising a cylindrical side wall with an open top end and a closed bottom end; the flange portion extending radially from the top end of the container portion, wherein a plurality of threaded holes are spaced around the flange;

a cylindrical cover manufactured to cover the top end of the container portion, wherein the cylindrical cover comprises a plurality of holes positioned to align with the holes of the flange portion; a series of fasteners to secure the cylindrical cover to the cylindrical light fixture base can; and a threaded insert applied to the plurality of threaded holes wherein the threaded insert comprises: a top end and a bottom end; a cylindrical wall running between the top and bottom ends of the threaded insert; a lip integral with the cylindrical wall proximate the bottom end of the threaded insert, wherein the lip has a diameter greater than the diameter of the cylindrical wall; an opening proximate the top end of the threaded insert and extending throughout the cylindrical wall to form a cavity between the interior of the cylindrical wall and the lip; a plurality of protrusions extending radially from the exterior of the cylindrical wall; a tapered portion proximate the end of the cylindrical wall opposite the lip, wherein the cavity extending throughout the cylindrical wall and tapered portion has a substantially uniform radius and the exterior of the tapered portion forms an angle with the plane of the cylindrical wall; and wherein the interior of the cylindrical wall is threaded.

11. The airfield light fixture apparatus of claim 10, wherein the flange comprises a top face and a bottom face and the threaded insert is applied to the plurality of threaded holes such that the lip is flush with the bottom face of the flange.

12. The airfield light fixture apparatus of claim 11, wherein the plurality of protrusions comprise a plurality of splines wherein the base of the spline extending from the cylindrical wall is greater in thickness than the top of the spline.

13. The airfield light fixture apparatus of claim 12, wherein the diameter of the plurality of splines is greater than the diameter of the plurality of threaded holes.

14. The airfield light fixture apparatus of claim 13, wherein the splines extend up to approximately 0.036 in. from the cylindrical wall.

15. The airfield light fixture apparatus of claim 13, wherein the fasteners are secured within the cavity of the threaded insert.

16. The airfield light fixture apparatus of claim 10, wherein the tapered portion forms a five degree angle with the plane of the cylindrical wall.

17. A method of repairing threading of an airfield light fixture comprising the steps of: removing a plurality of fasteners from an airfield light fixture apparatus comprising a cover, a base can having a flange which comprises a plurality of threaded holes, and the plurality of fasteners; removing the cover of the airfield light fixture apparatus; inspecting the threaded holes within the flange portion of the light fixture apparatus; clearing the threaded holes of any debris; drilling a plurality of threaded inserts into the plurality of threaded holes within the flange portion of the airfield light fixture; replacing the cover of the airfield light fixture apparatus; and reapplying the fasteners to the cover of the airfield light fixture apparatus.

18. The method of claim 17, wherein the plurality of threaded inserts applied to the plurality of threaded holes comprise: a top end and a bottom end; a cylindrical wall running between the top and bottom ends of the threaded insert; a lip integral with the cylindrical wall proximate the bottom end of the threaded insert, wherein the lip has a diameter greater than the diameter of the cylindrical wall; an opening proximate the top end of the threaded insert and extending throughout the cylindrical wall to form a cavity between the interior of the cylindrical wall and the lip; a plurality of protrusions extending radially from the exterior of the cylindrical wall; and a tapered portion proximate the end of the cylindrical wall opposite the lip, wherein the cavity extending throughout the cylindrical wall and tapered portion has a substantially uniform radius and the exterior of the tapered portion forms an angle with the plane of the cylindrical wall.

19. The method of claim 18 wherein the step of drilling a plurality of threaded inserts into the plurality of threaded holes within the flange portion of the airfield light fixture further comprises: aligning the top end of the threaded insert with the plurality of threaded holes along the bottom face

of the flange; and providing sufficient force to wedge the threaded insert into the threaded holes within the flange.
